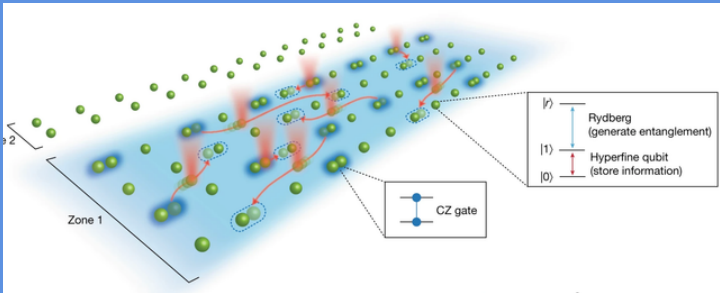


QUANTUM COMPUTING WITH NEUTRAL ATOMS

Kaili Murphy, Matty Flores, Hunter Wright, Karsten Hakes

WHAT IT IS

A gate-based QC system:



Each qubit is a single rubidium atom

Due to the use of neutral atoms, qubits are stable and have relatively long coherence times

“Room Temperature”: cooling needs are lower because lasers do the work instead of cryogenic cooling systems

HOW IT WORKS

Trapping & cooling:

Atoms held in optical tweezers and cooled near absolute zero

Qubit control:

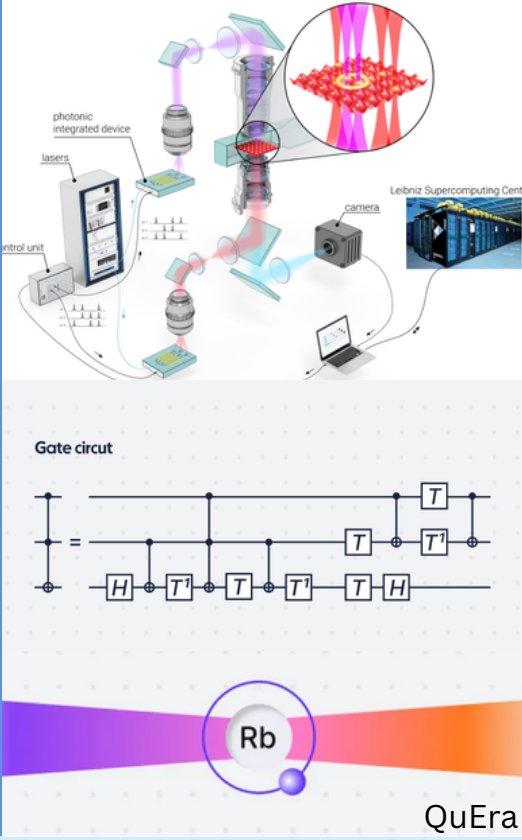
Energy levels encode information, manipulated by laser pulses

Rydberg interactions:

Enable entanglement between atoms

Quantum circuits:

Laser control allows gate operations and circuit execution



WHY IT MATTERS

- **Scalable & flexible:** Supports hundreds to thousands of qubits arranged and reconfigured with lasers
- **Expandable architecture:** Easier to scale than many other qubit technologies
- **Long coherence times:** Preserves quantum information for complex computations
- **High controllability:** Precise manipulation improves reliability
- **Real-world applications:** Useful for materials science, drug discovery, optimization, and cryptography



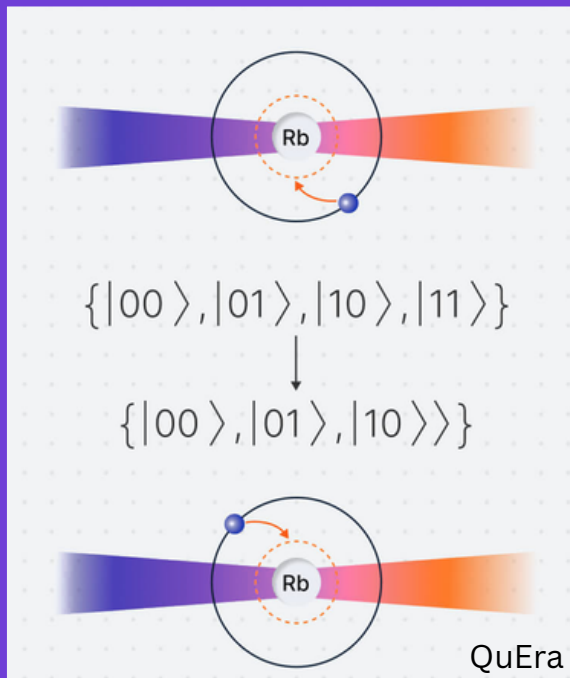
PROS AND CONS

Pros:

- Identical qubits
- Large-scale arrays
- Room temperature operation
- Implementation of conditional quantum logic and two-qubit gates. (QuEra)

Cons:

- Difficulty in controlling individual atoms and manipulating them



KEY PLAYERS

- QuEra Computing
- Pasqal
- Atom Computing
- Infleqtion
- planQC



CURRENT STATUS

EXPANSION

Quantum Computers with as little as 10000–20000 qubits (CalTech)

ERROR CORRECTION

CalTech experiment with 6000 qubits (CalTech) error corrections done in milliseconds, without disrupting the data

NEW COMPANIES

Google AI expanding to both Superconducting and Neutral Atoms (H. Neven)

SPEED

The biggest downfall is currently the speed, only measured in milliseconds